SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5

Code of Conduct

I K Sumanth Kumar Reddy (192425222) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K Sumanth

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 and Type-2 Hypervisors for a Cloud Datacenter

Answer

A **hypervisor** is software that creates and manages Virtual Machines (VMs).

Type-1 Hypervisor (Bare-Metal)

- Runs directly on the physical hardware.
- Does not require a host operating system.
- Examples: VMware ESXi, Microsoft Hyper-V, Xen.
- Provides high performance and strong security.

Type-2 Hypervisor (Hosted)

- Runs on top of a host operating system.
- Easier to install and use.
- Examples: VirtualBox and VMware Workstation.
- Suitable for testing and development.

Comparison

Feature	**Type-1**	**Type-2**
Performance	High	Moderate
Security	High	Lower
Speed	Faster	Slower
Resource Usage	Efficient	More Overhead

Justification

For a cloud datacenter hosting **mission-critical applications**, **Type-1 Hypervisor** is the better choice because it provides:

- Better performance
- Higher security
- Greater reliability
- Efficient resource utilization
- Enterprise-level management features

Hence, Type-1 hypervisors are widely used in cloud data centers.

2. Virtualization Architecture for Workload Isolation

Answer
Virtualization allows multiple virtual machines to run on a single physical server while keeping them isolated.
Workload Isolation
The finance, healthcare, and student portal applications run in separate VMs. Each VM has:

- Independent operating system
- Separate memory
- Separate storage
- Virtual CPU
- Separate virtual network

The hypervisor prevents one VM from accessing another VM's resources.
Major Attack Surfaces
1. Hypervisor Attack

- If the hypervisor is compromised, attackers may gain access to multiple VMs.

2. VM Escape

- A malicious VM escapes its environment and attacks other VMs or the hypervisor.

Conclusion
Virtualization provides secure workload isolation, efficient resource sharing, and improved security. Proper patching, access control, and monitoring help reduce attack risks.

3. VM Host Utilization Analysis

Answer
Given Data

- CPU Utilization = **92%**
- Memory Utilization = **88%**
- Storage I/O Latency = **35 ms**
- VM Boot Time = **170 seconds**

Analysis
CPU (92%)

- CPU is overloaded.
- Applications may become slow.

Memory (88%)

- Memory usage is very high.
- System may start swapping memory.

Storage Latency (35 ms)

- High latency indicates slow disk performance.
- VMs take longer to access data.

Boot Time (170 s)

- Long boot time is caused by CPU, memory, and storage bottlenecks.

Corrective Actions

- Increase CPU resources.
- Upgrade RAM.
- Use SSD or NVMe storage.
- Balance workloads across multiple hosts.
- Remove unnecessary VMs.

Conclusion
The major bottlenecks are CPU overload, high memory usage, and slow storage. Resource upgrades and load balancing will improve VM performance.

4. Software Defined Datacenter (SDDC) vs Traditional Datacenter

Answer
A **Software Defined Datacenter (SDDC)** virtualizes compute, storage, and networking, allowing infrastructure to be managed through software.

Compute Virtualization

- Multiple virtual machines run on one physical server.
- Resources can be allocated dynamically.

Storage Virtualization

- Combines multiple storage devices into one logical storage pool.
- Improves storage flexibility and utilization.

Network Virtualization

- Creates virtual networks independent of physical hardware.
- Enables faster network configuration.

Advantages over Traditional Datacenter

- Faster deployment
- Better scalability
- Automated resource management
- Improved flexibility
- Reduced operational cost

Conclusion
Compared to traditional datacenters, SDDC offers greater agility through automation, virtualization, and centralized management of compute, storage, and networking resources.

5. Multi-Cloud Federation Identity Management

Answer
In a multi-cloud environment, different cloud providers may use different identity systems, causing authentication and access issues.

Problem Analysis

- Different user accounts across providers
- Inconsistent access permissions
- Authentication failures
- Security risks

Proposed Solution
Use **Identity Federation** with:

- Single Sign-On (SSO)
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)

Protocols such as **SAML**, **OAuth 2.0**, or **OpenID Connect (OIDC)** allow users to authenticate once and securely access multiple cloud providers.

Privacy and Security

- Encrypt data using AES-256.
- Protect communication using TLS/SSL.
- Maintain audit logs for monitoring.
- Apply least-privilege access through RBAC.

Conclusion
Identity federation improves security, simplifies user management, and enables secure access across multiple cloud providers while maintaining privacy and compliance.